

LEIBSTADT, Switzerland

WMO: 066660

Lat: 47.5972N

Lon: 8.1883E

Elev: 341

StdP: 97.29

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.1	(c) -5.6	(d) -12.3	(e) 1.4	(f) -6.9	(g) -8.7	(h) 1.9	(i) -4.2	(j) 10.2	(k) 5.8	(l) 9.1	(m) 5.8	(n) 2.5	(o) 60	(p) 0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	32.0	20.7	29.9	20.2	27.8	19.4	21.9	29.1	21.0	27.9	20.1	26.1	2.6	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.7	15.0	24.0	18.8	14.2	22.8	18.1	13.6	22.1	65.4	29.2	62.1	28.2	59.2	26.1	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.6	(b) 6.4	(c) 5.4	(e) -10.7	(f) 34.9	(g) 3.6	(h) 1.4	(i) -13.3	(j) 35.9	(k) -15.4	(l) 36.8	(m) -17.4	(n) 37.6	(o) -20.0	(p) 38.6		
(a) 7.6	(b) 6.4	(c) 5.4	(e) -11.1	(f) 24.2	(g) 3.5	(h) 1.3	(i) -13.6	(j) 25.2	(k) -15.6	(l) 25.9	(m) -17.6	(n) 26.7	(o) -20.1	(p) 27.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	1.4	1.9	6.4	10.7	14.2	18.3	20.2	19.7	15.6	11.0	5.9	2.1
	DBStd	7.54	3.62	4.04	3.39	3.49	3.53	3.29	3.12	3.19	3.05	3.49	3.42	3.36
	HDD10.0	1056	268	226	117	34	6	0	0	0	1	32	129	244
	HDD18.3	2994	526	459	369	228	136	42	17	22	92	227	374	502
	CDD10.0	1297	1	0	6	56	134	248	315	302	168	63	6	0
	CDD18.3	195	0	0	0	0	7	39	74	66	9	0	0	0
	CDH23.3	1996	0	0	0	15	99	436	758	592	94	1	0	0
	CDH26.7	682	0	0	0	1	16	144	289	213	18	0	0	0
Wind	WSAvg	1.8	2.0	2.0	2.1	1.9	2.0	1.8	1.8	1.5	1.5	1.5	1.6	1.8
Precipitation	PrecAvg	1326	118	109	98	92	115	113	115	107	91	106	120	143
	PrecMax	1825	297	286	355	188	196	213	220	239	184	220	291	331
	PrecMin	948	14	23	21	4	35	43	30	19	36	6	4	1
	PrecStd	220	73	70	74	49	44	46	45	44	40	54	68	81
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	15.2	20.0	25.8	29.3	33.2	35.1	33.9	28.9	22.4	17.3	12.1
		MCWB	10.0	9.6	11.5	15.4	19.4	22.2	21.8	20.6	19.3	16.1	13.0	10.3
	2%	DB	9.3	11.7	17.8	22.8	26.0	30.5	32.2	31.4	26.2	20.0	14.5	10.2
		MCWB	7.6	7.7	10.5	13.6	16.9	20.9	20.2	20.9	18.5	15.5	11.7	9.2
	5%	DB	7.6	9.5	15.4	20.9	23.9	28.0	29.9	29.2	23.5	17.7	12.4	8.4
		MCWB	6.4	6.9	9.6	13.4	16.1	19.8	19.8	20.4	17.6	14.5	10.5	7.3
	10%	DB	6.0	7.2	12.9	18.5	21.4	25.7	27.7	26.7	21.4	15.9	10.4	6.8
		MCWB	5.1	5.3	8.6	12.3	15.3	18.6	19.2	19.5	16.6	13.9	9.2	6.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.8	10.2	12.5	15.8	20.0	23.8	22.7	23.0	20.5	17.2	13.5	11.1
		MCDB	12.3	13.4	17.8	23.2	26.3	30.8	31.8	30.0	26.4	20.5	16.3	11.2
	2%	WB	7.9	8.3	11.2	14.6	18.4	21.6	21.5	21.8	19.2	16.1	12.0	9.2
		MCDB	9.0	11.3	16.4	21.2	23.7	28.7	29.3	28.4	24.3	18.5	14.1	10.1
	5%	WB	6.6	6.9	10.0	13.5	16.9	20.4	20.6	20.9	18.2	15.2	10.6	7.6
		MCDB	7.5	9.3	14.3	19.4	21.8	26.3	27.6	27.5	22.3	17.1	12.2	8.2
	10%	WB	5.2	5.4	9.0	12.5	15.8	19.3	19.8	20.1	17.2	14.1	9.4	6.0
		MCDB	5.9	7.1	12.5	17.9	20.3	24.5	25.8	25.6	20.3	15.6	10.3	6.8
Mean Daily Temperature Range	5% DB	MDBR	4.7	6.7	9.9	11.4	10.6	11.1	11.8	11.4	10.4	8.1	5.5	4.6
		MCDBR	7.1	11.4	15.5	16.6	15.2	15.9	16.4	15.6	14.1	11.1	8.9	6.9
		MCWBR	5.5	7.6	8.8	8.3	7.2	7.1	5.8	6.2	7.0	6.8	6.0	5.6
	5% WB	MCDBR	6.9	10.7	13.5	15.1	13.2	14.7	14.3	14.3	12.4	9.9	8.0	6.6
		MCWBR	5.5	7.5	7.9	7.8	6.6	6.9	5.6	6.0	6.3	6.4	5.9	5.7
Clear-Sky Solar Irradiance	taub	0.310	0.338	0.375	0.406	0.418	0.422	0.422	0.415	0.390	0.368	0.333	0.306	
	taud	2.429	2.341	2.281	2.226	2.237	2.272	2.265	2.299	2.347	2.385	2.435	2.432	
	Ebn at Noon	770	817	838	845	847	844	837	827	813	770	732	726	
	Edh at Noon	67	90	112	130	134	130	129	120	104	86	67	60	
All-Sky Solar Radiation	RadAvg	1.10	1.92	3.14	4.32	5.02	5.80	5.56	4.85	3.64	2.27	1.23	0.90	
	RadStd	0.13	0.28	0.44	0.62	0.51	0.53	0.51	0.49	0.37	0.23	0.15	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (114 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+0.73	+0.42	+0.33	N/A	N/A	+55	+14

Nomenclature: See separate page